

Attention-Based Multi-Instance Learning for Whole-Slide Image Classification

(2024)

Problem

Classifying gigapixel histopathology slides without requiring patch-level labels.

Method

A multi-instance learning framework with attention-based pooling over slide patches.

Dataset

CAMELYON16 lymph node metastasis dataset.

Evaluation

AUC and tumor localization accuracy.

Results

The attention-MIL model achieved 0.91 AUC, with attention maps loosely correlated to tumor regions.

Limitations

Attention weights correlate imperfectly with tumor regions as confirmed by pathologist review.